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StreamYOLO26: Wide resolution multi-object detection for Autonomous Vehicles

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Abstract:	Autonomous Vehicles (AVs) increasingly rely on Advanced Driver-Assistance Systems (ADAS) to promote safe and reliable driving in urban and highway environments, particularly at Levels 3 to 5 of driving automation, where robust perception is essential for safe decision-making. Vision-based object detection identifies and localizes surrounding vehicles, pedestrians, and traffic infrastructure, reducing reliance on multiple sensors. However, conventional detectors neglect temporal information, while existing YOLO-based streaming detectors may struggle with consistent object detection in dynamic driving scenarios. To address these limitations, this study introduces StreamYOLO26, a real-time streaming object detection framework built on the latest YOLO26 backbone that exploits temporal information from consecutive frames to improve detection accuracy and consistency. The framework incorporates a C3k2-based Dual Flow Process (DFP) module to fuse temporal features from consecutive frames while retaining the YOLO26 detection pipeline, including its DFL-free detection head, small-target-aware label assignment (sTAL), CIoU and L1 localization losses, binary cross-entropy classification loss, and the Muon with Stochastic Gradient Descent (MuSGD) optimizer. Training follows a two-stage strategy, initially fine-tuning the newly introduced parameters with pretrained components frozen, followed by end-to-end fine-tuning on the Argoverse-HD dataset for direct comparison with existing models. The proposed large model achieves a stream Average Precision (sAP) of 39.2%, with an additional computational cost of 69.18 GFLOPs while reducing latency by 4.2 ms during online evaluation. In contrast, the proposed extra-large model achieves the highest sAP of 42.0% during offline evaluation, demonstrating improved detection accuracy at the expense of increased computational requirements.

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Cover Letter

StreamYOLO26: Wide resolution multi-object detection for Autonomous Vehicles

Nihar Ranjan Swain, Anand J. Kulkarni, Mandar S Sapre, Puskaraj D. Sonawwanay, Sanjay Patil

To

The Editor-in-Chief

Engineering Applications of Artificial Intelligence

Dear Editor,

We are pleased to submit our original research manuscript entitled “StreamYOLO26: Wide resolution multi-object detection for Autonomous Vehicles” for consideration for publication in Engineering Applications of Artificial Intelligence. In this study, a streaming object detection framework is proposed using pretrained YOLO26 models with shared weights. The framework takes the current and previous frames as input to predict bounding boxes for the next frame, aiming to reduce inference latency and support real-time perception in autonomous driving scenarios.

The proposed nano, small, medium, large, and extra-large models were trained and evaluated on the Argoverse-HD dataset. The validation results demonstrate that the proposed next-frame prediction models outperform existing real-time object detection methods. Additional testing shows that the models reliably detect large and clearly visible objects; however, detection performance remains challenging for occluded objects and those at long distances. These findings highlight the potential of the proposed approach for real-time autonomous vehicle perception, while indicating that training with additional and diverse datasets could further improve its robustness and generalisation capability.

The manuscript is well suited to Engineering Applications of Artificial Intelligence because it presents a deep learning-based computer vision framework that integrates convolutional feature extraction and attention-based feature fusion to enhance object detection accuracy and support real-time perception in autonomous driving applications. We confirm that this manuscript is original, has not been published, and is not under consideration elsewhere. All authors have approved the content and its submission to your esteemed journal. There are no conflicts of interest to disclose, and the study adheres to all applicable ethical guidelines.

Kindly consider our work for publication in your Journal.

Sincerely,

Nihar Ranjan Swain

Anand J. Kulkarni

Mandar S Sapre

Puskaraj D. Sonawwanay

Sanjay Patil

Manuscript without author details Click here to view linked References

StreamYOLO26: Wide resolution multi-object detection for Autonomous Vehicles

ABSTRACT

Autonomous Vehicles (AVs) increasingly rely on Advanced Driver-Assistance Systems (ADAS) to promote safe and reliable driving in urban and highway environments, particularly at Levels 3 to 5 of driving automation, where robust perception is essential for safe decision-making. Object detection plays a vital role in environmental perception, particularly for collision avoidance. To reduce dependence on multiple sensor technologies, vision-based object detection enables the identification and localization of surrounding vehicles, pedestrians, and traffic infrastructure. However, conventional detectors process frames independently and therefore fail to exploit temporal cues from preceding frames, while existing streaming detectors based on earlier YOLO architectures may struggle to consistently detect all objects in dynamic driving scenarios. To address these limitations, this study introduces StreamYOLO26, a real-time streaming object detection framework built on the latest YOLO26 backbone that exploits temporal information from consecutive frames to improve detection accuracy and consistency. The framework incorporates a C3k2-based Dual Flow Process (DFP) module to fuse temporal features from consecutive frames while retaining the YOLO26 detection pipeline, including its DFL-free detection head, small-target-aware label assignment (sTAL), CIoU and L1 localization losses, binary cross-entropy classification loss, and the Muon with Stochastic Gradient Descent (MuSGD) optimizer. Training follows a two-stage strategy, initially fine-tuning the newly introduced parameters with pretrained components frozen, followed by end-to-end fine-tuning on the Argoverse-HD dataset for direct comparison with existing models. The proposed large model achieves a stream Average Precision (sAP) of 39.2%, with an additional computational cost of 69.18 GFLOPs while reducing latency by 4.2 ms during online evaluation. In contrast, the proposed extra-large model achieves the highest sAP of 42.0% during offline evaluation, demonstrating improved detection accuracy at the expense of increased computational requirements.

Keywords:

Autonomous Vehicle, Streaming Object Detection, Convolution, Attention, Computer Vision.

1 Introduction

The rapid advancement of AVs has established deep learning as a key perception technology that complements conventional sensor systems. Environmental perception enables the estimation of surrounding objects, their depth, and velocity, supporting safe decision-making during driving. Object detection is a fundamental perception task for identifying and localizing surrounding objects, thereby reducing the risk of collisions. However, existing state-of-the-art (SOTA) detectors are largely developed for offline benchmarks and fail to satisfy the real-time accuracy, latency, and robustness requirements of AV applications (Lakshmanan et al., 2022; Swain et al., 2024).

Object detection methods are broadly classified into two-stage detectors, such as R-CNN (Lin et al., 2014) which generate region proposals before object classification and localisation, and one-stage detectors, such as YOLO (Redmon et al., 2016) which perform both tasks in a single pass. The prominent YOLO series has progressively advanced in accuracy, efficiency, and real-time performance.YOLOv4 (Bochkovskiy et al., 2020) introduced a three-stage learning architecture for object detection, a backbone for hierarchical and discriminative feature extraction, a neck for multi-scale feature aggregation and fusion, and a detection head for predicting object classes and bounding-box coordinates. It also developed Bag of Freebies (BoF) to improve accuracy without increasing model complexity, and Bag of Specials (BoS) to enhance accuracy through additional architectural components. YOLOv8 (Reis et al., 2023) introduced C2f and Spatial Pyramid Pooling- Fast (SPPF) block in its backbone and PAN-based neck and adopted an anchor-free detection paradigm, predicting bounding-box distances from the anchor points without anchor boxes. The detection head removed the objectness branch and used a task-aligned assigner to enhance classification and localization through weighted box loss and Distribution Focal Loss (DFL). YOLOv10 (A. Wang et al., 2024) introduced Compact Inverted Block (CIB) and Partial Self-Attention (PSA) modules in the backbone and neck, with NMS-free training using one-to-many and one-to-one label assignments and only the one-to-one head during inference. YOLOv11(Khanam & Hussain, 2024) introduced the C3k2 block in the backbone and placed the C2PSA block after the SPPF layer to enhance feature representation. YOLOv13 (Lei et al., 2025) introduces depth-wise convolution through DS-Conv and DS-C3K2 blocks in the backbone, along with HyperACE module that forwards all PAN blocks to enhance feature representation and information

aggregation. YOLO26 (Jocher et al., 2026) is a recent SOTA based on YOLOv11 architecture with the MuSGD optimizer, Progressive Loss, which integrates one-to-one and one-to-many prediction strategies, sTAL label assigner, DFL-free regression and end-to-end, NMS-free inference for efficient detection.



(a) (b) (c) (d)

Fig. 1 Representative images from the Argoverse-HD training set under diverse driving conditions: (a) Clear daytime, (b) Morning, (c) Rainy weather, and (d) Nighttime.

Online streaming detection overcomes conventional detection by adopting sAP to evaluate continuous video detection using Mask R-CNN with a ResNet backbone (M. Li et al., 2020). StreamYOLO(Yang et al., 2022) used a pretrained YOLOX backbone and neck with shared weights, which exploited temporal information from the previous and current frames through a DFP to predict bounding boxes for the next frame. It also introduces a velocity term to improve temporal object prediction during streaming inference. The Chanakya (Ghosh et al., n.d.) model employed deep reinforcement learning for streaming object detection but achieved lower sAP than SOTA models. LongShortNet (C. Li et al., 2023) uses a YOLOX backbone and Long-Short Fusion module(LSFM) to fuse temporal features from multiple preceding frames for subsequent frame detection. DAMO-StreamNet (He et al., 2023) adopted a temporal modelling strategy similar to LongShortNet while introducing a Dynamic Receptive Field Feature Pyramid Network (DRFPN) with ELAN blocks to overcome the limitations of YOLOX's fixed spatial receptive field in capturing temporally varying features. Furthermore, the model employed a dual-path architecture incorporating the LSFM to effectively aggregate temporal information across consecutive frames. IS-YOLO (Tang et al., 2026) follows the StreamYOLO and advances the DFP with channel implicit knowledge through channel-wise Gram matrices to capture implicit relationships between current and previous frame features after FPN extraction.

The Argoverse-HD (M. Li et al., 2020) dataset, as shown in Fig. 1 is selected for its diverse 65 training and 24 validation video sequences captured under varying weather and time conditions. In accordance with the initial experiments, a novel Dilated Neighbourhood Attention-based block is incorporated into C3k2, alongside Global Attention and HybridHyperACE modules combining Graph Convolution and Graph Attention, followed by the YOLOv13 architecture. However, the model has a higher parameter count and exhibits vanishing-gradient issues after three to four epochs. Its major limitation is the substantial dataset requirement for effective training from scratch. To address these challenges, the backbone and neck are initialized with pretrained YOLO26 weights, and the experiments are conducted across the Nano, Small, Medium, Large, and Extra-Large model variants. Following the experimental settings of StreamYOLO and IS-YOLO, the proposed approach uses the previous and current frames to predict bounding boxes for the next frame, thereby maintaining a streaming inference setting consistent with real-time deployment. The DFP module follows the StreamYOLO framework for temporal feature fusion. After fusion, the features are processed by the C3k2 block for efficient feature extraction and then combined with the current-frame features to incorporate more recent information. The YOLOv8 detection head then predicts bounding boxes and class scores for the eight object classes in the Argoverse-HD dataset. The STAL strategy is used to assign positive samples from the predicted results. The training objective comprises a temporal-weighted loss for the bounding-box and L1 losses, while the classification loss is calculated over all predicted samples to represent object presence. Finally, the model is trained using the MuSGD optimizer with a cosine learning-rate scheduler.

The major contribution of this work is described below.

- The development of StreamYOLO26, which employs the YOLO26 backbone for improved detection accuracy and uses separate weights for the backbone and neck processing of current and previous frames instead of sharing pretrained backbone weights.
- A modified DFP is introduced to fuse temporal features after feature extraction. Features from the current and previous frames are concatenated and processed through an attention-free C3k2 block, and the resulting fused features are combined with the current-frame features.

- The loss function combines the fundamental formulation of StreamYOLO with the YOLO26 loss framework to establish the StreamYOLO26 loss function, incorporating temporal weighting for bounding-box regression and L1 losses, together with the sTAL label assignment strategy
- In contrast to StreamYOLO and IS-YOLO, the proposed model uses multi-stage training at each iteration and relies only on one-to-many supervision rather than the Progressive Loss strategy.
- NMS is also applied during inference to provide reliable detection in challenging autonomous-driving scenarios, including heavy traffic, occluded objects, distant targets, and crowded scenes.

The paper is organised as follows. Section 2 presents the overall streaming object detection pipeline, key module developments, detection head types, label assignment strategies, and types of IoU. Section 3 describes the proposed model architecture, its mathematical formulations, and the overall training workflow. Section 4 details the experimental setup. Section 5 presents and analyses the experimental results. Finally, Section 6 discusses the limitations of the proposed model and future research directions.

Table 1
Types of Head

Category	Head	Principle
Anchor Based	Coupled Head(Lv et al., 2023)	Predicts bounding-box offsets relative to predefined anchor boxes, with shared feature representations for classification and regression, using one-to-many label assignment and NMS-based filtering during inference
Anchor free	Decoupled Head(Lv et al., 2023)	Predicts object locations and bounding boxes without predefined anchors, using decoupled classification and regression branches, one-to-many assignment, and NMS-based inference.
Distribution based	DFL Head(Reis et al., 2023)	Predicts bounding-box distance distributions using DFL, one-to-many assignment, and NMS-based inference
Transformer	Transformer query-based head (Y. Wang et al., 2022)	Uses transformer attention and object queries for one-to-one prediction without NMS.
Distribution free	DFL-free Head(Jocher et al., 2026)	Combines one-to-many and one-to-one branches with DFL-free RegMax=1, using NMS according to the prediction strategy

Table 2
Comparison of Label Assignment Methods

Assignment paradigm	Method	Positive-sample selection mechanism
Fixed / heuristic	IoU-threshold assignment(Redmon et al., 2016)	Anchors with IoU above the positive threshold are assigned as positives; low-IoU anchors as negatives
Geometry-based / anchor-free	FCOS (Z. Tian et al., 2019)	Feature points whose centres lie inside the GT box are considered candidates
Adaptive	ATSS (S. Zhang et al., 2020)	Candidate anchors are selected according to an adaptive threshold derived from the mean and standard deviation of their IoUs
	AutoAssign (Zhu et al., 2020)	Automatically learns spatial and scale-dependent positive-sample weights using classification and localization information
Quality-aware	QFL-based assignment(X. Li et al., 2020)	Localization quality is incorporated into the classification target
	Varifocal assignment(H. Zhang et al., 2021)	IoU-derived quality scores are used as continuous classification targets
	IoU-aware assignment (Kim & Lee, 2020)	Predictions with higher localization quality receive greater supervision
One-to-one matching	Hungarian matching (Carion et al., 2020)	A unique prediction is optimally matched to each GT using the Hungarian method
	Bipartite matching (Carion et al., 2020)	Global matching produces a unique assignment between predictions and GTs
Global optimization	OTA (Ge et al., 2021)	Prediction–GT assignments are globally optimized under transport constraints
Dynamic matching	SimOTA (Yang et al., 2022)	A dynamically determined number of lowest-cost candidates is selected for each GT
	Weighted SimOTA (Tang et al., 2026)	It enhances SimOTA by weighting dynamically assigned positive samples according to their classification confidence and localization quality.
Task-aligned	TAL(Khanam & Hussain, 2024)	Candidates are ranked using a task-alignment metric combining classification confidence and localization quality
	sTAL (Jocher et al., 2026)	Enhances TAL by selecting positive samples based on task alignment and assigning normalized weights according to classification confidence and localization quality.

2 Related Studies

An object detection model broadly consists of three components: the backbone extracts meaningful image features, the neck fuses multi-scale features for detecting objects of different sizes, and the detection head predicts bounding boxes and object classes. The major backbone and neck modules are: Cross Stage Partial Network (CSPNet) splits feature into two paths and partially aggregates them to reduce redundancy and computational.

CSPDenseNet applies the same CSPNet principle to dense features. Efficient Layer Aggregation Network (ELAN) uses multiple computational paths and structured layer aggregation to improve gradient flow and feature reuse (C. Y. Wang et al., 2023). Generalized ELAN (GELAN) combines CSP and ELAN-based feature aggregation with structural re-parameterization for efficient feature extraction (C. Y. Wang et al., 2025). C3k replaces standard bottleneck convolutions with configurable kernel-size operations to enlarge the receptive field. C3k2 is an efficient CSP-style block that uses internal bottleneck, C3k, or attention modules to enhance feature extraction (Khanam & Hussain, 2024).

C2f splits features and passes intermediate bottleneck outputs to concatenation for richer gradient flow. Spatial Pyramid Pooling (SPP) applies pooling at multiple receptive-field sizes and concatenates the resulting features to capture multi-scale contextual information. Nevertheless, SPPF sequentially applies pooling operations to capture multiple receptive fields with lower computational cost than conventional SPP (Reis et al., 2023). HyperACE adaptively aggregates multi-scale features through hypergraph convolution to capture global high-order correlations and C3k-based processing to model local spatial relationships, enabling effective interactions among neighbouring and cross-scale features while preserving shortcut information (Lei et al., 2025). C2PSA integrates attention into a CSP-style block to enhance spatially informative feature representation (Khanam & Hussain, 2024). Table 1 presents a comparative analysis of the major types of detection heads used in object detection models. The proposed model employs a DFL-free one-to-many detection head with NMS-based inference. Table 2 presents a comparative analysis of label assignment methods used to select positive samples based on their correspondence with ground-truth bounding boxes. The proposed model employs the STAL label assigner, a SOTA label assignment strategy for positive-sample selection. Table 3 presents a comparative analysis of IoU-based functions used for bounding-box regression loss. The proposed model employs CIoU due to its effective localization of large objects, which is particularly relevant to autonomous driving scenarios.

Table 3
Comparison of IoU Metrics Used in the Object Loss Function for Object Detection

Method	Mechanism
IoU - Intersection over Union (Redmon et al., 2016)	Overlap between predicted and ground truth boxes
GIoU - Generalized IoU (Rezatofighi et al., 2019)	Overlap + enclosing box penalty
DIoU - Distance IoU (Zheng et al., 2020)	Overlap + center-point distance
CIoU - Complete IoU (Zheng et al., 2020)	Overlap + center distance + aspect ratio
EIoU - Efficient IoU (Y. F. Zhang et al., 2022)	Overlap + center distance + width/height differences
SIoU - Scylla IoU (Gevorgyan, 2022)	Overlap + distance + angle + shape
α-IoU (HE et al., 2021)	Generalized IoU optimization using α
WIoU -Wise IoU (Tong et al., 2023)	Dynamic weighting based on prediction quality
Focal-CIoU (Y. F. Zhang et al., 2022)	CIoU with focal weighting
Shape-IoU (H. Zhang & Zhang, 2023)	IoU with explicit shape awareness
Inner-IoU (H. Zhang et al., 2023)	Overlap of scaled inner boxes + localization sensitivity
MPDIoU - Minimum Point Distance IoU (Ma & Xu, 2023)	Minimum-point distance between bounding boxes
NWD-IoU - Normalized Wassertein Distance with IoU (J. Wang et al., 2021)	Overlap + normalized Wasserstein distance
PIoU – Penalized IoU (Chen et al., 2020)	Overlap + localization penalty

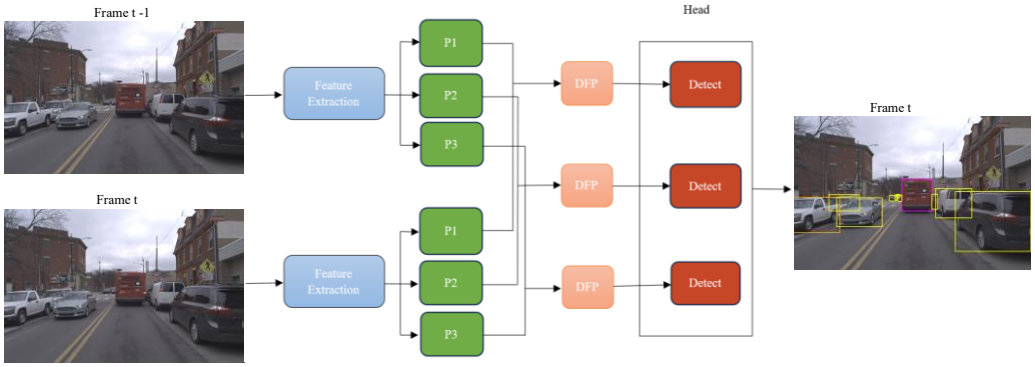


Fig. 2 Proposed Temporal Streaming Object Detection Framework

3 Methods

The complete architecture of the proposed model is illustrated in Fig. 2. The model predicts next-frame bounding boxes using the current and previous frames as inputs. The YOLO26 backbone and neck are employed for multi-level feature extraction, while a C3K2 layer is incorporated into the DFP network. The detection head follows the

YOLO26 design with eight-class predictions. The loss function is further modified using temporal weights and the STAL assigner. Finally, the model is trained using the MuSGD optimizer with a cosine learning-rate scheduler.

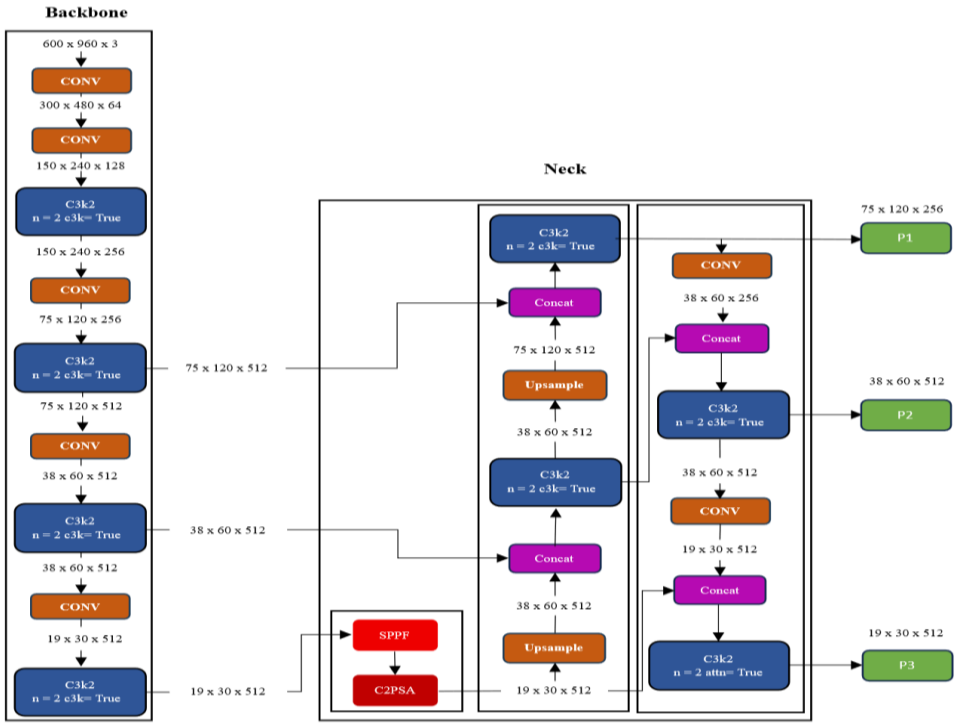


Fig. 3 Feature Extraction for Large Model

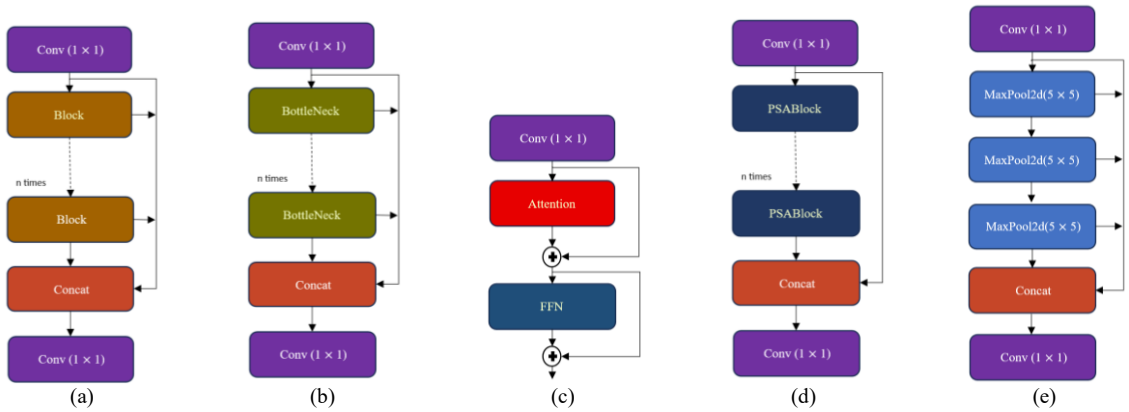


Fig. 4 Block diagrams of the feature extraction modules: (a) C3k2, (b) C3k, (c) PSABlock, (d) C2PSA, and (e) SPPF.

3.1 Feature Extraction

The complete feature extraction process for the 600 × 960 × 3 input in the large StreamYOLO26 model, using independent weights for the current and previous frames, is shown in Fig. 3 each input frame x_t for this experiment is represented as a tensor of H × W × C, where H, W and C denote height, width, and number of colour channels, respectively. The input is first passed through the backbone, where the initial block consists of convolutional layers with Batch Normalization and SiLU activation for high-level feature extraction. The C3k2 block is highly inspired by the CSP network. Within its block structure, it uses either a Bottleneck consisting of simple convolutional layers, a C3k block, or a combination of Bottleneck and PSA blocks, as shown in Fig. 4(a), Fig. 4(b) and Fig. 4(c). The neck passes the three extracted feature maps through to generate three rich multi-level feature representations. The final feature map extracted by the backbone is passed through the SPPF, a pooling-based module illustrated in Fig. 4(e). Its output is then processed by C2PSA, a CSP-based attention module shown in Fig. 4(d). The neck consists of Conv and C3K2 blocks for multi-level feature processing. The final layer of each model incorporates an attention module to enhance feature representation for larger objects commonly

considered in real-time object detection. The final feature extraction process produces three feature maps, P_1, P_2, and P_3, which are used for subsequent detection.

3.2 Dual Flow Process

The DFP fuses feature maps from the current and previous frames. Following StreamYOLO, the feature maps are first reduced by half along the channel dimension and then concatenated. Although the spatial implicit model addresses the attention limitation of ISYOLO, it misses more bounding boxes than StreamYOLO. This may result from the Gram matrix compressing the features into a single representation before restoring the original feature-map dimensions, potentially weakening spatial relationships between the current and previous frames. In addition, the model showed limited compatibility with the MuSGD optimizer and other approach like the attention mechanism, which introduce approximately 10x higher computational cost. Therefore, a C3k2 block is adopted to process the concatenated features without attention, providing a simpler and computationally efficient approach for spatial feature extraction. Finally, the refined fused features are added to the current-frame feature map to obtain the final feature representation, as expressed in Eq. (3.1), where F_t and F_{t-1} denote the feature maps of the current and previous frames, respectively, as shown Fig. 5.

$$F_{\text{fused}} = F_t + \text{C3k2}(\text{Concat}[\text{Conv}(F_t), \text{Conv}(F_{t-1})]) \tag{3.1}$$

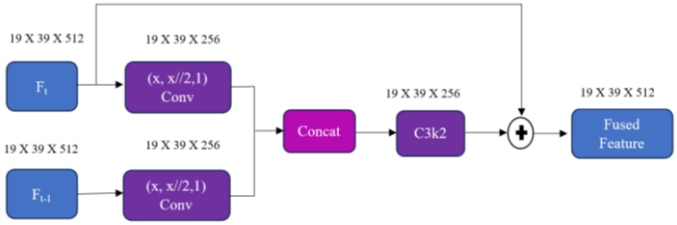


Fig. 5 Dual Flow Process

3.3 Detection Head

The detection head is divided into two parts as shown in Fig. 6, with the first predicting the bounding-box distance distributions from the anchor centre. It consists of two convolutional blocks, with the channel dimensions defined in Eq. (3.2), where C_2 is the minimum number of channels among the P_1, P_2, and P_3 and R represents the number of RegMax bins. Class prediction is treated as a multilabel classification task using a sequence of DWConv and Conv blocks. The first block maps the input channels to C_3 colour channels as shown in Eq. (3.3), where nc is the number of classes. The final prediction layers for both class and bounding-box distance use only Conv2d.

$$C_2 = \max\left(16, \left\lfloor \frac{C_{\text{in}}}{4} \right\rfloor, 4R\right) \tag{3.2}$$

$$C_3 = \max(C_{\text{in}}, 80, \min(\text{nc}, 10)) \tag{3.3}$$

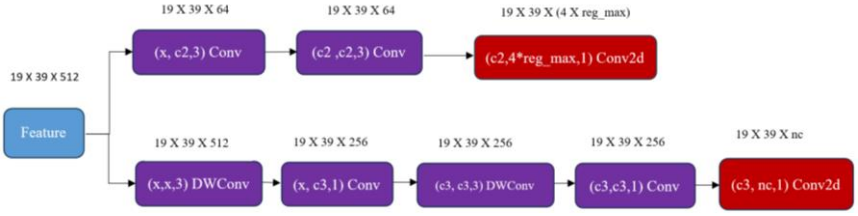


Fig. 6. Detection Head

3.4 Label assigner and Loss Function

The IoU metric is widely used to measure the overlap between ground-truth and predicted bounding boxes; however, it becomes zero for non-overlapping objects. In this study, CIoU is used for all object categories. Eq. (3.4)

-(3.7) present the IoU and CIoU formulations. Where d is the Euclidean distance between the predicted and ground truth box, c is the smallest enclosing box diagonal length. w and h are the predicted bounding-box width and

height; $w_{gt}$and h_{gt} are the ground-truth width and height; v is the aspect-ratio consistency, and alpha is the adaptive weight for aspect ratio.

$$\mathrm{IoU} = \frac{\mathrm{Intersection}}{\mathrm{Union}} \tag{3.4}$$

$$\mathrm{CIoU} = \mathrm{IoU} - \frac{d^2}{c^2} - \alpha v \tag{3.5}$$

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w_{gt}}{h_{gt}} - \arctan\frac{w}{h}\right) \tag{3.6}$$

$$\alpha = \frac{v}{(1-\mathrm{IOU}) + v} \tag{3.7}$$

The temporal small-target-aware label assignment strategy improves the temporal robustness of the streaming object detection model by filtering positive and negative samples. Following the prediction of bounding boxes and confidence scores, anchor points are generated across all multi-scale feature maps. These anchor points serve as reference locations for assigning each predicted bounding box to its corresponding spatial location. The first step is to identify positive anchor centres located within the ground-truth bounding boxes. Subsequently, an alignment metric is computed for each predicted bounding box, as defined in Eq. (3.8). Where S_{a,c_g} denotes the predicted classification probability to the ground truth, $CIOU_{g,a}$ denotes localisation overlap between the predicted bounding box and ground truth bounding box. α and β are the control contributions of classification and localisation quality. The assigner then selects the top-(k) alignment metrics for each ground-truth box. If an anchor is assigned to multiple ground-truth boxes, the assignment is resolved by selecting the ground-truth box with the highest overlap. Finally, it returns positive target boxes, target labels and normalised target score.

$$A_{ga} = \left(S_{a,c_g}\right)^{\alpha} \left[\max\left(\mathrm{CIoU}_{g,a}, 0\right)\right]^{\beta} \tag{3.8}$$

The loss function comprises box loss, classification loss, and DFL. With RegMax = 1, the DFL is replaced by L1 loss (Mean Absolute Error). A temporal weight is constructed from the ground-truth bounding boxes of the current and next frames, forming a temporal IoU matrix. For each ground-truth object in the next frame, the maximum IoU with the ground-truth objects in the current frame is selected. The resulting temporal weight is defined in Eq. (3.9).

$$w_{i,a}^{\mathrm{temp}} = \frac{1}{\left\{\max\left[\max_k \mathrm{IoU}\left(B_{i,g}^{\mathrm{next}}, B_{i,k}^{\mathrm{Curr}}\right),\ 0.2\right]\right\}^{\alpha} + \epsilon} \tag{3.9}$$

As the objectness loss is excluded from the total loss, a target-score weight is assigned to each positive anchor, calculated as the sum of the classification target scores across classes, as defined in Eq. (3.10). This target-score weight is then combined with the CIoU loss to obtain the per-anchor localization loss, as expressed in Eq. (3.11). Subsequently, Eq. (3.12) defines the temporal box weight by normalizing the per-anchor losses, thereby incorporating the spatial correspondence of objects across consecutive frames. Finally, the box loss is computed by weighting the per-anchor loss with the temporal box weight and normalizing it by the total target-score sum, as given in Eq. (3.13).

$$w_a^{\mathrm{cls}} = \sum_{c=1}^{C} t_{a,c} \tag{3.10}$$

$$\ell_a^{\mathrm{box}} = \sum_{a\in\mathcal{F}} w_a^{\mathrm{cls}} \left[1 - \mathrm{CIoU}\left(B_a, \hat{B}_a\right)\right] \tag{3.11}$$

$$q_a^{\mathrm{box}} = \frac{w_a^{\mathrm{temp}} \ell_a^{\mathrm{box}}}{\sum_{j\in\mathcal{F}} w_j^{\mathrm{temp}} \ell_j^{\mathrm{box}} + \epsilon} \tag{3.12}$$

$$\mathcal{L}_{\mathrm{box}} = \frac{\sum_{a\in\mathcal{F}} q_a^{\mathrm{box}} \ell_a^{\mathrm{box}}}{S_{\mathrm{target}} + \epsilon} \tag{3.13}$$

The L1 loss (Mean Absolute Error), defined in Eq. (3.14) measures the absolute difference between the ground-truth and predicted distances from the anchor center. Similar to the box loss, a temporal weight is incorporated into the L1 loss, as defined in Eq. (3.15) to account for temporal correspondence across consecutive frames. The final L1 loss is obtained by weighting the per-anchor L1 loss with the temporal weight and normalizing it by the total target-score sum, as given in Eq. (3.16).

$$\ell_a^{\mathrm{L1}} = \frac{w_a^{\mathrm{cls}}}{4} \sum_{k\in\{l,t,r,b\}} \left| d_{a,k}^{*} - \hat{d}_{a,k} \right| \tag{3.14}$$

$$q_a^{\mathrm{L1}} = \frac{w_a^{\mathrm{temp}} \sum_{a\in\mathcal{F}} \ell\, a^{\mathrm{L1}}}{\sum j \in \mathcal{F}\, w_j^{\mathrm{temp}} \ell_j^{\mathrm{L1}} + \epsilon} \tag{3.15}$$

$$\mathcal{L}_{\mathrm{L1}} = \frac{\sum_{a\in\mathcal{F}} q_a^{\mathrm{L1}}\, \ell_a^{\mathrm{L1}}}{S_{\mathrm{target}} + \epsilon} \tag{3.16}$$

The classification loss is formulated using Binary Cross-Entropy (BCE), treating object detection as a multi-label classification problem. Within the assigner, positive anchors whose centers lie inside the ground-truth bounding boxes are assigned a target score of 1, while the remaining anchors are assigned 0. However, the classification loss is computed over all predicted scores, where a higher score indicates a greater likelihood of object presence. The BCE and classification loss formulations are presented in Eq. (3.17) and Eq. (3.18).

$$\mathcal{L}_{\mathrm{BCE}}(y, \hat{y}) = -[y \log(\hat{y}) + (1-y)\log(1-\hat{y})] \tag{3.17}$$

$$\mathcal{L}_{\mathrm{Class}} = \frac{\sum_{a\in\mathcal{F}} \mathrm{BCE}\,(p_a, t_a)}{S_{\mathrm{target}} + \epsilon} \tag{3.18}$$

The final loss function is the combination of box loss, class loss and L1 loss, which is written in Eq. (3.19) with its coefficients, which regulate the importance during training.

$$\mathcal{L}_{\mathrm{Loss}} = \lambda_1 \mathcal{L}_{\mathrm{box}} + \lambda_2 \mathcal{L}_{\mathrm{class}} + \lambda_3 \mathcal{L}_{\mathrm{L1}} \tag{3.19}$$

3.5 Optimizer

The optimizer is essential for deep learning, as it iteratively updates model parameters using gradients obtained through backpropagation to minimize the loss function. Several widely adopted optimization algorithms are available for training deep learning models, including Stochastic Gradient Descent (SGD), Adaptive Gradient Algorithm (Adagrad), Root Mean Square Propagation (RMSProp), Nesterov Accelerated Gradient (NAG), Adaptive Moment Estimation (Adam), and Adam with Decoupled Weight Decay (AdamW) (Hemmati et al., 2026; Y. Tian et al., 2023). In this experiment, the MuSGD optimizer is used, which is a combination of SGD and the Muon Optimizer (Jocher et al., 2026). Gradient Descent (GD) computes gradients over the entire dataset and updates model parameters after each complete pass, whereas Stochastic Gradient Descent (SGD) updates the parameters after each iteration using an individual sample or mini-batch (Y. Tian et al., 2023). Eq. (3.20) describes the weight update rule used in SGD.

$$w_t = w_{t-1} - n\nabla L(f(x_i, w_{t-1}), y_i) \tag{3.20}$$

Similar to Adam, Muon (K. Jordan et al., 2024) maintains a running momentum estimate, as expressed in Eq. (3.21). The resulting momentum matrix is subsequently orthogonalized using the Newton–Schulz iteration, as described in Eq. (3.22). Finally, the Muon parameter update with weight decay is formulated in Eq. (3.23).

$$\beta_t = \mu\beta_{t-1} - \nabla L_t(w_{t-1}) \tag{3.21}$$

$$O_t = \mathrm{NewtonSchulz}(\beta_t) \tag{3.22}$$

$$w_t = w_{t-1} - n(O_t + \lambda w_{t-1}) \tag{3.23}$$

The MuSGD uses both SGD momentum and orthogonalized momentum updates as shown in the Eq. (3.24). Inside the MUSGD of SGD part if it uses Nesterov Momonetum it uses from eqution Eq. (3.25).

$$v_t = \beta v_{t-1} + n\nabla L_t(\theta_{t-1}) \tag{3.24}$$

$$w_t = w_{t-1} - n\{\alpha_M O_t + \alpha_S[\nabla L_t(\theta_{t-1}) + \lambda w_{t-1} + \beta v_t]\} \tag{3.25}$$

3.6 Mean Average Precision

Precision shows how good is model predict among all predicated positive detection as shown in Eq. (3.26) . Recall suggest how good model is among all ground truth shown in Eq. (3.27). Eq. (3.28) is the interpolated precision which helps to non-increasing precision labels. Eq. (3.29) calculates the average precision, Where $P(\tilde{r})$ is the

precision at recall $\tilde{r}$ and r is the recall level. The COCO metric calculates AP by averaging interpolated precision over 101 recall points. Eq. (3.30) defines the mean Average Precision (mAP), obtained by averaging AP across IoU thresholds ranging from 0.50 to 0.95 in increments of 0.05 and across all object classes (Padilla et al., 2021). The streaming Average Precision (sAP) metric was adopted as the primary evaluation metric for this experiment (M. Li et al., 2020). Similar to mean Average Precision (mAP) (Lin et al., 2014), sAP measures the average precision across IoU thresholds ranging from 0.50 to 0.95 for small, moderate, and large objects. However, unlike conventional mAP, sAP calculates the average precision for next-frame prediction with next-frame ground truth.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3.26}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3.27}$$

$$P_{\text{interp}}(r) = \max_{\tilde{r} \geq r} P(\tilde{r}) \tag{3.28}$$

$$AP = \frac{1}{101} \sum_{j=0}^{100} P_{\text{interp}}\left(\frac{j}{100}\right) \tag{3.29}$$

$$\text{mAP}_{\text{COCO}} = \frac{1}{10C} \sum_{c=1}^{C} \sum_{t=0}^{9} A\,P_c(0.50) \tag{3.30}$$

4 Experiment

4.1 Dataset

The experiments were conducted using the Argoverse-HD high-frame-rate (30 FPS) dataset, which comprises urban outdoor scenes captured in two U.S. cities, Pittsburgh and Arlington (Chang et al., 2019). The dataset contains eight object classes: 'Person', 'Bicycle', 'Car', 'Motorcycle', 'Bus', 'Truck', 'Traffic Light', and 'Stop Sign'. For training and validation, the centre RGB camera images and their corresponding annotations were used, with the dataset split into 24 videos for training and 15 videos for validation, respectively (M. Li et al., 2020).

4.2 Model Evaluation

The offline and online evaluation methods assess the performance of the proposed model. The offline evaluation procedure compares predictions from the current and previous frames with the next frame's ground-truth annotations. The image ID associated with each prediction is stored to ensure accurate matching with the corresponding ground-truth annotations in the validation dataset. Nevertheless, online evaluation behaves like continuous streaming evaluation, where the model adopts an online setup. Initially, the model predicts the bounding boxes, and the results are stored with timestamps. From the next frame onward, the first frame acts as the previous frame, and the incoming frame acts as the current frame for prediction until the sequence is complete in the validation dataset. The evaluation is then calculated by matching each prediction with the ground-truth annotations corresponding to the same timestamp. This temporal matching provides a more realistic assessment of detector performance in continuous video streams like AVs driving scenarios. As a result, the online sAP score may be lower than the offline sAP during validation because it considers temporal consistency and accounts for missed or unmatched frames. Nevertheless, this evaluation is particularly relevant to autonomous vehicle applications, where reliable and temporally consistent object detection across continuous video frames is essential for safe perception and decision-making. The Floating-Point Operations (FLOPs) and the number of parameters were also calculated to evaluate the computational complexity of the models.

4.3 Implementation Details

All experiments were implemented using the PyTorch framework in Python and were conducted on a workstation equipped with two NVIDIA RTX A4500 GPUs for both model training and streaming evaluation. The training loop is followed by a YOLO26-style training loop with modifications to streaming object detection, where the previous and current frames are used to predict next frame bounding boxes (Yang et al., 2022). Training the model from scratch is computationally expensive and requires extensive hyperparameter tuning. Therefore, this study utilizes the pretrained YOLO26 (n/s/m/l/x) model weights along with the optimized optimizer and loss function hyperparameters provided by the original implementation. Furthermore, the standard MuSGD optimizer was used during training to maintain consistency with the YOLO26 baseline. Unlike StreamYOLO and IS-YOLO, the proposed model employs multi-scale training, where the input image size is randomly varied at each training

iteration while maintaining a stride of 32. The input resolution ranges from a minimum of 416×640 pixels to a maximum of 736×1152 pixels. To maintain geometric consistency in streaming object detection, only horizontal flipping augmentation was applied. For each experiment, the matched pretrained weights were frozen for the first five epochs to train the detection head and Dual Flow module. Subsequently, all layers were unfrozen, and the entire model was trained for an additional 3 epochs to enable all network layers to fully learn the characteristics of the ArgoverseHD datasets. Batch size is varied among 2,4,8,16 and 32 for extra-large, large, medium, small and nano, respectively.

5 Results and Analysis

5.1 Ablation Studies

All ablation experiments were conducted using the small model to maintain computational efficiency. The experiments were performed as fine-tuning experiments based on COCO pre-trained weights(Jocher et al., 2026) for 15 epochs.

5.1.1 Ablation of the Optimal Model

The first ablation study was conducted on the backbone architecture to identify the optimal model configuration. The first experiment involved extracting features from the previous and current frames, followed by feature fusion using a simple convolution-based addition mechanism. Second, an additional C3k2 block was applied after feature fusion. In the third experiment, global attention was incorporated during the feature extraction process. Although the global attention model has a higher number of parameters and incorporates an additional attention layer, its sAP decreased because the attention mechanism disrupted the pretrained feature representations during training. Applying the C3k2 block after feature fusion improves feature extraction efficiency. The sAP results obtained from all three experiments are presented in Table 4. The C3k2-based model achieved an 11.3% higher sAP than the baseline model and a 24.1% higher sAP than the global attention model. Therefore, the C3k2-based architecture was selected for further experimentation.

Table 4
Ablation for model architecture selection

Model Name	sAP	sAP_{50}	sAP_{75}	sAP_s	sAP_m	sAP_l
Streamyolo26s-without_c3k2	30.9	48.9	31.6	13.1	31.5	55.7
Streamyolo26s-with_c3k2	**31.2**	**48.4**	**32.2**	**14.5**	**31.1**	**53.6**
Streamyolo26s-with_global_attn	24.3	37.7	25.3	10.5	22.9	42.5

5.1.2 Ablation of the optimal RegMax number in detection head

The RegMax parameter determines the number of discrete distributions used to represent the four bounding-box distances ((l,t,r,b)) relative to the anchor centre. To evaluate its influence on detection performance, experiments were conducted using RegMax values of 1, 8, and 32, with the corresponding results reported in Table 5. Among the evaluated configurations, RegMax = 1 achieved the highest sAP, outperforming RegMax = 8 and RegMax = 32 by 7.4% and 0.9%, respectively. However, these results should be interpreted in the context of the adopted training configurations. The RegMax = 1 model was initialised with the complete pretrained YOLO26 weights, whereas the RegMax = 8 and 32 configurations were trained from scratch because their modified detection heads were incompatible with the original pretrained weights. This difference in initialisation resulted in distinct optimization behaviour during training, particularly in the DFL loss, and may have contributed to the observed variation in sAP. Therefore, while RegMax = 1 demonstrated the best performance under the evaluated experimental settings, the improvement cannot be attributed solely to the RegMax configuration.

Table 5
Ablation for optimal RegMax in detection head

Model Name	sAP	sAP_{50}	sAP_{75}	sAP_s	sAP_m	sAP_l
Streamyolo26s-regmax=1	30.9	18.9	31.6	13.1	31.5	55.7
Streamyolo26s-regmax=8	29	48.7	29.3	13.3	29.6	47.8
Streamyolo26s-regmax=16	**31.1**	**50.5**	**31.3**	**13.5**	**32.0**	**53.5**
Streamyolo26sregmax=32	28.0	48.8	27.7	12.2	30.7	46.6

5.1.3 Searching for training approach

The final ablation study evaluates the training strategy. The model parameters were frozen for the first five epochs using pretrained weights and then fully unfrozen for the subsequent 15 epochs to adapt the model to the target

dataset. Three configurations were evaluated: RegMax = 1 without freezing, RegMax = 1 with freezing, and RegMax = 16 with freezing. For RegMax = 16, YOLO26 pretrained weights were used for the backbone and neck, while YOLO11 pretrained weights were used for the detection head due to its compatibility with RegMax = 16. The results are presented in Table 6. The RegMax = 1 with freezing configuration achieved the best performance, with 8.33% higher sAP than RegMax = 1 without freezing and 4.64% higher sAP than RegMax = 16 with freezing.

Table 6
Ablation for optimal training approach

Model Name	sAP	sAP_{50}	sAP_{75}	sAP_s	sAP_m	sAP_l
Streamyolo26s-c3k2-regmax=1	31.2	48.4	32.2	14.5	31.1	53.6
Streamyolo26s-c3k2-regmax=1_freeze	**33.8**	**53.6**	**34.9**	**13.9**	**34.8**	**62.2**
Streamyolo26s-c3k2-regmax=16_freeze	32.3	52.0	32.7	13.9	33.6	55.6

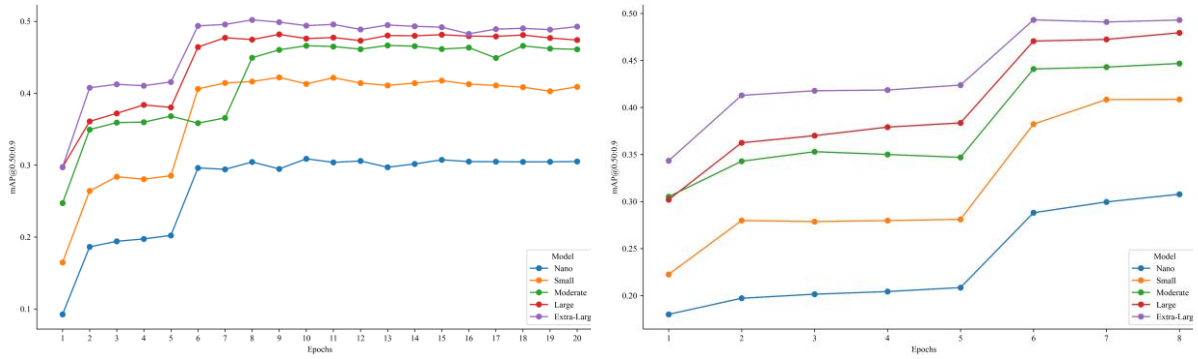
5.2 Training Analysis

The final model configuration, established through ablation studies, incorporates the C3k2 block after current–previous frame feature fusion, RegMax = 1, and a two-stage training strategy. The model is first trained with frozen parameters for five epochs, followed by full-network training for 15 epochs, as shown in Fig. 7(a), and for three epochs, as shown in Fig. 7(b). The experiments were conducted using five model variants: Nano (N), Small (S), Medium (M), Large (L), and Extra-Large (X).

The five-epoch freezing stage allowed the task-specific parameters to adapt while preserving the pretrained features. After unfreezing, full-network fine-tuning led to a sharp increase in mAP, and the models converged within 3–4 epochs before gradually reaching a performance plateau. As shown in Fig. 7(a), the mAP sometimes dropped after reaching its peak during training and then increased again. As a result, the final evaluation performance decreased by 1–2%. Therefore, all experiments were conducted for up to eight epochs, as shown in Fig. 7(b). Fig. 7 compares the training performance of all models across epochs using mAP@0.50:0.95, evaluated at IoU thresholds from 0.50 to 0.95 in increments of 0.05. The COCO-pretrained models initially achieved low mAP on the target dataset. A comparison of the training and evaluation metrics in Table 7 shows that the Extra-Large (X) model achieves the highest overall performance.

Table 7
Training evaluation of all models using map

Model Name	Precision	Recall	mAP50	mAP50-95	params(M)	FLOPs(B)
StreamYOLO26-N (Ours)	67.24	39.5	50.86	30.75	2.899	15.98
StreamYOLO26-S (Ours)	75.4	49.36	61.89	40.88	11.47	62.16
StreamYOLO26-M (Ours)	74.37	52.75	64.13	44.46	23.78	201.57
StreamYOLO26-L (Ours)	79.32	55.77	67.54	47.74	29.65	261.34
StreamYOLO26-X (Ours)	79.94	**57.91**	**70.65**	49.34	66.62	585.61



(a) (b)

Fig. 7 Comparison of epochs with mAP@0.50:0.95 for all training models (a) 20 epochs (b) 8 epochs

5.3 Comparison with SOTA

This section compares our model with all other streaming detection models with AgroverseHD datasets for both offline and online evaluation.

5.3.1 Offline evaluation

Fig 8 shows that, overall, the proposed models outperform both baselines across most object classes, demonstrating the effectiveness of the unshared backbone and neck with the proposed feature-fusion approach.

For the Small model, only the Bicycle class shows lower sAP, while all other classes outperform both baselines. For the Medium model, the Bicycle class and Motorcycle remain lower than both baselines. For the Large model, Motorcycle shows lower sAP than the baselines. The relatively lower performance for Bicycle and Motorcycle may be attributed to their small image representation and greater occurrence at distant locations, which can increase the likelihood of missed detections.

Table *8* compares the proposed models with existing real-time streaming and non-real-time object detectors. For a fair comparison, all models use a consistent inference resolution of 600 × 960 for next-frame detection. The proposed approach introduces lightweight Nano, Small, Medium, Large, and Extra-Large variants that achieve 24.4%, 33.8%, 37.2%, 40%, and 42% sAP, respectively, outperforming their corresponding StreamYOLO and IS-YOLO counterparts. The Extra-Large model achieves the highest overall performance among the compared models, except for LongShort, which benefits from additional training data, and the model that ranks first using a higher resolution. Furthermore, the sAP75 values of 41.9% and 43.8% for the Large and Extra-Large models, respectively, demonstrate improved localization accuracy, particularly for small objects. Although the Extra-Large model achieves higher sAP, its FLOPs are almost twice those of the Large model, indicating that the Medium or Large model is more suitable for real-time implementation. Adding more historical frames may lead to a reduction in model performance; therefore, this study is restricted to only the current and previous frames for next frame prediction.

5.3.2 Online evaluation

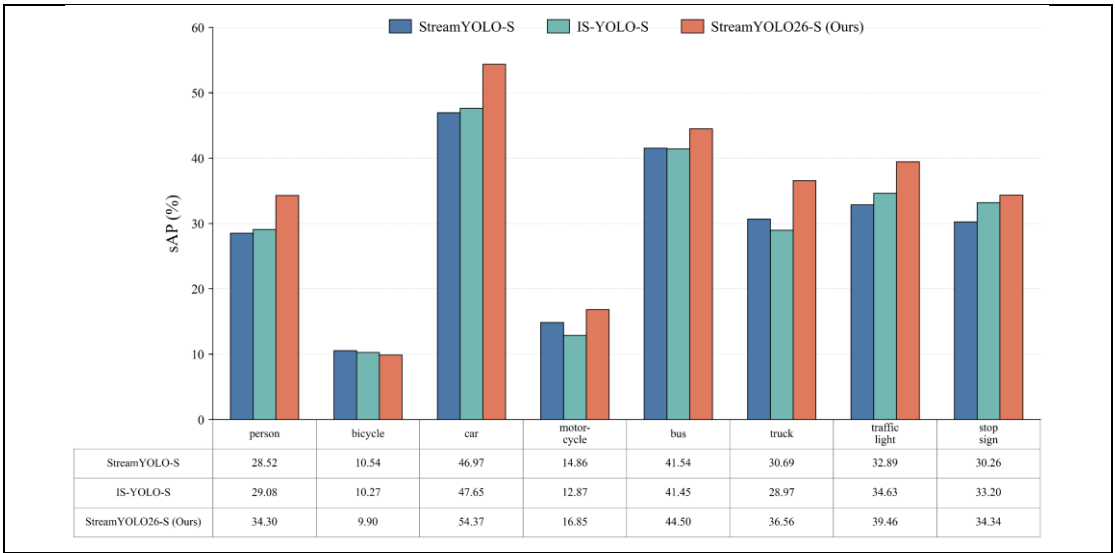
The online evaluation follows the StreamYOLO evaluation protocol and is performed using two NVIDIA RTX A4500 GPUs at an input resolution of 600 × 960 pixels. To emulate real-world autonomous driving conditions, the dataset is organized into video sequences and streamed at 30 FPS to enable sequential next-frame object detection. During inference, the model processes each frame sequentially and stores the predictions corresponding to their respective timestamps. For evaluation, the predicted results are aligned with the ground-truth annotations based on timestamps, enabling the computation of all evaluation metrics, including class-wise sAP.

Table *9* compares the proposed models with StreamYOLO and IS-YOLO in terms of online detection performance and latency. The results report the mean latency for each model; however, the relatively high standard deviation indicates considerable variation in inference latency during online evaluation. Although matching 30 FPS is good practice, the large model is suitable for comparison with existing models. In the experiment, online evaluation was conducted using models ranging from Nano to Large, excluding the Extra-Large model, which requires higher computational resources and fails to produce sAP in this setup. A slight drop in online sAP compared with offline sAP is observed, as expected, because detection times may increase for some frames. In comparison with existing studies, the proposed Large model achieves 39.2% sAP with a mean latency of 27.3 ms, which is even lower than that of the StreamYOLO model at 30 FPS, with an additional 69.18 GFLOPs.The proposed model's lower latency and higher sAP suggest that it delivers faster predictions with higher accuracy. By adopting the modern YOLO26 backbone and a DFP with a C3k2 block, the model is suitable for real-time AV deployment.

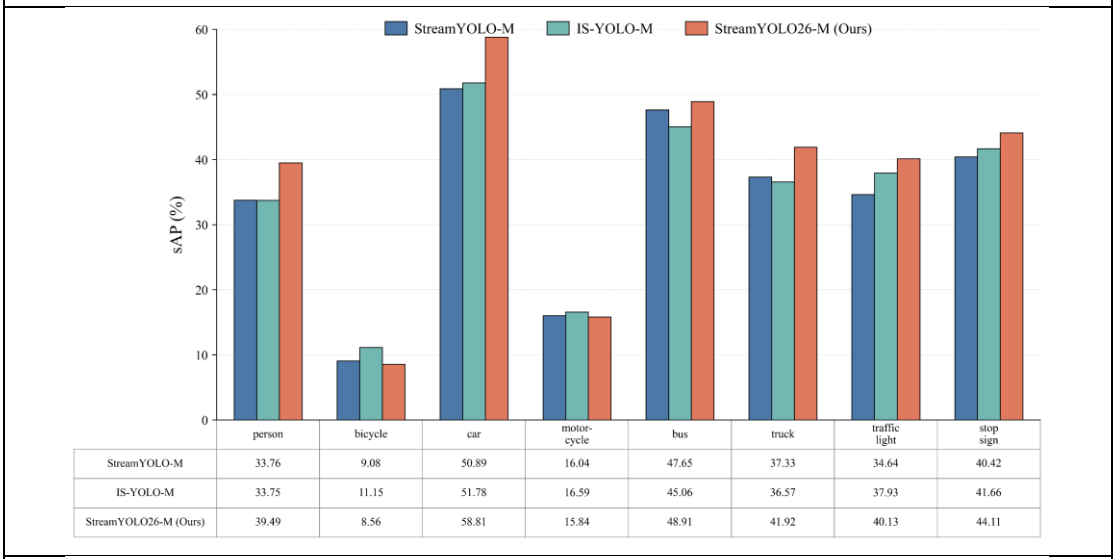
5.4 Visual Analysis

In this section, the proposed model is compared with StreamYOLO and IS-YOLO. Fig. 9 shows a comparative prediction with existing model, the proposed large model shows more accurate object detection than existing methods. The model boxes are closer to the ground truth boxes, which shows superior detection compared to both models. It indicates that increased computation and modern architecture show more streaming detection. The model also produces less false positives. Despite the more accurate prediction, the proposed model like IS-YOLO and YOLO26 model misses occluded and large-distance objects because the models are given an attention layer only for the depth layer to focus more on large features in the neck layer.

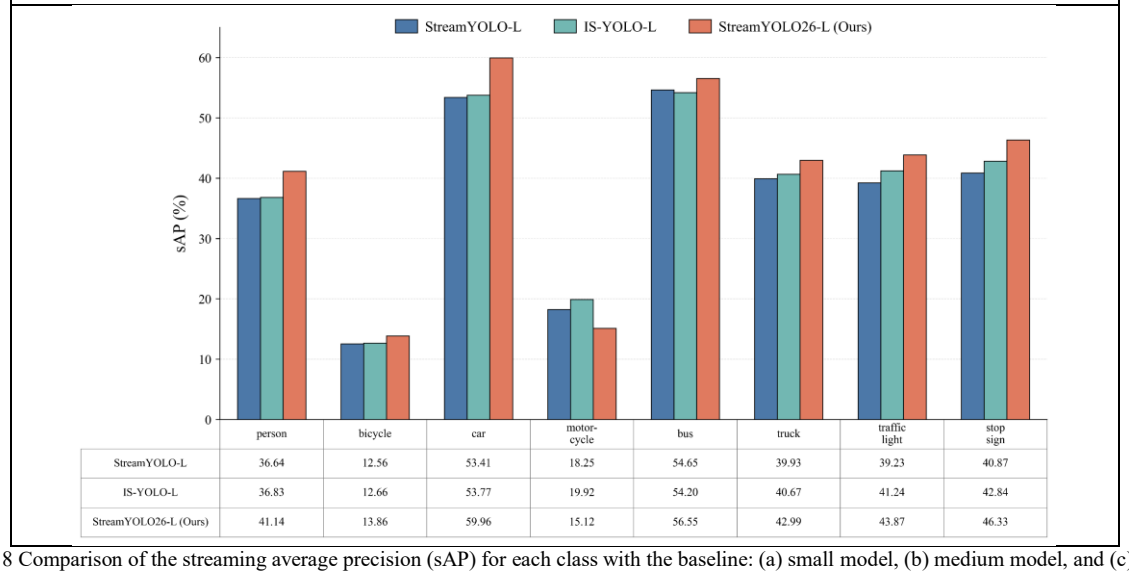
The large model was tested on all test-case images, as shown in Fig. 10, where the correctly detected objects in each test sequence are shown in red. However, overall, it misses distant and some occluded objects. In Fig. 10, the second image in the first row clearly shows that the model fails to detect all bicycles as their number increases. Interestingly, the model is still able to detect objects during night driving with high-beam lights and to detect large objects under rainy conditions. Overall, the model's predictions are satisfactory for handling objects at distances of approximately 100 m. The only flaw is observed in the third image of the second row, where the model incorrectly predicts some buildings as trucks.



	person	bicycle	car	motor-cycle	bus	truck	traffic light	stop sign
StreamYOLO-S	28.52	10.54	46.97	14.86	41.54	30.69	32.89	30.26
IS-YOLO-S	29.08	10.27	47.65	12.87	41.45	28.97	34.63	33.20
StreamYOLO26-S (Ours)	34.30	9.90	54.37	16.85	44.50	36.56	39.46	34.34



	person	bicycle	car	motor-cycle	bus	truck	traffic light	stop sign
StreamYOLO-M	33.76	9.08	50.89	16.04	47.65	37.33	34.64	40.42
IS-YOLO-M	33.75	11.15	51.78	16.59	45.06	36.57	37.93	41.66
StreamYOLO26-M (Ours)	39.49	8.56	58.81	15.84	48.91	41.92	40.13	44.11



	person	bicycle	car	motor-cycle	bus	truck	traffic light	stop sign
StreamYOLO-L	36.64	12.56	53.41	18.25	54.65	39.93	39.23	40.87
IS-YOLO-L	36.83	12.66	53.77	19.92	54.20	40.67	41.24	42.84
StreamYOLO26-L (Ours)	41.14	13.86	59.96	15.12	56.55	42.99	43.87	46.33

Fig. 8 Comparison of the streaming average precision (sAP) for each class with the baseline: (a) small model, (b) medium model, and (c) large model.

Table 8
Comparison with other state-of-the-art (SOTA) methods on the AgroverseHD dataset. “Size” denotes the shorter spatial dimension of the input image. * indicates that the model was trained using additional datasets, while † indicates the use of

additional historical data during training. Our method uses a validation input size of 600 × 960 with pretrained YOLO26 weights. Here, N, S, M, L, and X denote the Nano, Small, Medium, Large, and Extra-Large model variants, respectively.

Model Name	sAP	sAP_{50}	sAP_{75}	sAP_s	sAP_m	sAP_l
Non-real-time methods						
Streamer(size=900) (M. Li et al., 2020)	18.2	35.3	16.8	4.7	14.4	34.6
Streamer(size=600) (M. Li et al., 2020)	20.4	35.6	20.8	3.6	18.0	47.2
Streamer + AdaScale (Chin et al., 2019)	13.8	23.4	14.2	0.2	9.0	39.9
Chanakya (Ghosh et al., n.d.)	21.3	37.3	21.1	4.4	18.7	47.1
Real-time methods						
1st place* (size=1440) (S. Zhang et al., 2021)	40.2	68.9	39.4	21.5	42.9	53.9
2nd Place* (size=1200) (Gu et al., 2021)	33.2	58.6	30.9	13.3	31.9	40.0
Real-time methods + next frame prediction						
Streamyolo-S (Yang et al., 2022)	28.8	50.3	27.6	9.7	30.7	53.1
Streamyolo-M (Yang et al., 2022)	32.9	54.0	32.5	12.4	34.8	58.1
Streamyolo-L (Yang et al., 2022)	36.1	57.6	35.6	13.8	37.1	63.3
LongShortNet -S (C. Li et al., 2023)	29.8	50.4	29.5	11.0	30.6	52.8
LongShortNet -M (C. Li et al., 2023)	34.1	54.8	34.6	13.3	35.3	58.1
LongShortNet -L (C. Li et al., 2023)	37.1	57.8	37.7	15.2	37.3	63.8
LongShortNet - L† (C. Li et al., 2023)	42.7	65.4	45.6	23.9	44.8	61.8
DAMO-StreamNet-S† (He et al., 2023)	31.8	52.3	31.0	11.4	32.9	58.7
DAMO-StreamNet-M† (He et al., 2023)	35.7	56.7	35.9	14.5	36.3	63.3
DAMO-StreamNet-L† (He et al., 2023)	37.8	59.1	38.6	16.1	39.0	64.6
IS-YOLO-S (Tang et al., 2026)	29.8	49.1	30.1	11.6	30.2	52.4
IS-YOLO-M (Tang et al., 2026)	34.3	55.0	35.2	13.9	35.5	59.8
IS-YOLO-L (Tang et al., 2026)	37.8	58.4	38.2	15.6	38.0	65.7
StreamYOLO26-N (Ours)	24.4	42.1	24.2	7.6	25.2	45.6
StreamYOLO26-S (Ours)	33.8	53.6	34.9	13.9	34.8	62.2
StreamYOLO26-M (Ours)	37.2	55.8	39.8	17.1	38.7	63.0
StreamYOLO26-L (Ours)	40.0	59.2	41.9	18.7	40.6	64.4
StreamYOLO26-X (Ours)	**42.0**	**62.9**	**43.8**	**21.7**	**42.8**	**66.3**



Fig. 9 Real-time detection comparison with the existing model. The green boxes are ground truth boxes and red boxes predicted boxes

Table 9

Model Name	sAP	Latency Mean	Latency std	FLOPs
Streamyolo-L	36.9	28.1ms	1.6ms	192.15G
IS-YOLO-L	37.3	31.5ms	2.0ms	192.16G
Streamyolo26-N	24.3	17.9ms	9.71ms	15.98G
Streamyolo26-S	33.6	17.7ms	9.05ms	62.16G
Streamyolo26-M	37.0	19.2ms	8.37ms	201.57G
Streamyolo26-L	39.2	27.3ms	7.6ms	261.347G



Fig. 10 Predicted detections across test sequences under rainy conditions, beam illumination, and long-distance scenarios, highlighting robust detection performance while illustrating limitations in occluded scenes and a false-positive truck classification for a building.

6 Limitations and Future Directions

The proposed model demonstrates competitive performance compared with existing SOTA methods; however, further improvements are still required. The current model operates at a latency close to the real-time requirement of 30 FPS; further reductions in computational latency are essential to support autonomous vehicles operating at higher speeds. The model still struggles to detect distant and small objects; therefore, future work will investigate more efficient and powerful feature-extraction architectures to improve detection performance, including C3k2–ELAN combinations, lightweight attention mechanisms such as Flash Attention, graph convolution, and graph attention, to enhance feature representation, particularly for long-range and small-object detection. Further improvements will also explore advanced detection heads and loss functions to strengthen overall detection performance. Multi-stream fusion and real-time processing can reduce detection accuracy by approximately 25%, highlighting the need for more efficient and robust streaming models.

Future research will focus on 3D streaming object detection for autonomous vehicles to provide richer spatial-temporal information for safer decision-making and collision avoidance. Large-scale datasets covering diverse

and adverse weather conditions will also be required to evaluate model robustness and generalization. Furthermore, emerging vision-language models offer opportunities to develop advanced streaming detection methods and unified perception frameworks for autonomous vehicles.

7 Conclusion

The proposed StreamYOLO26 framework employs independent YOLO26 backbones for the current and previous frames to exploit temporal information for next-frame object detection. The DFP module integrated with the C3k2 block enhances temporal feature fusion while reducing the computational overhead associated with attention-based mechanisms and spatial implicit modules. This design supports real-time inference while mitigating the risk of missing objects. The framework further incorporates a temporal cue-based streaming loss derived from the YOLOv8 detection loss, together with the sTAL label assigner, to improve detection performance across consecutive frames. The primary challenge is that training a large-scale model from scratch may result in suboptimal detection performance. Therefore, all experiments adopt a two-stage training strategy, initially freezing the pretrained weights and subsequently fine-tuning the entire network to maximise sAP. Experimental results demonstrate that the proposed StreamYOLO26 variants outperform the evaluated existing methods. In particular, the extra-large variant achieves 42.0% sAP under offline evaluation, whereas the large variant attains 39.2% sAP under online evaluation. The large model achieves good overall detection performance on the test dataset, with occasional false-positive detections of buildings as trucks. However, it struggles to detect objects at long distances and under significant occlusion.

Although the extra-large model achieves a higher sAP in offline evaluation, it incurs nearly twice the computational cost of the large model. Therefore, the large and medium models are more suitable for real-time implementation. The findings highlight two essential stages in improving streaming object detection: first, extracting discriminative spatial features using an effective object detection backbone and, second, integrating temporal information through an efficient feature fusion strategy. The combination of these stages is crucial for improving detection accuracy while maintaining computational efficiency in real-time autonomous driving applications. Incorporating additional datasets can further improve the model's detection performance and generalisation capability for streaming object detection.

Data availability

The data will be made available upon request. The implementation will be available on GitHub.

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StreamYOLO26: Wide resolution multi-object detection for Autonomous Vehicles

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ABSTRACT

Autonomous Vehicles (AVs) increasingly rely on Advanced Driver-Assistance Systems (ADAS) to promote safe and reliable driving in urban and highway environments, particularly at Levels 3 to 5 of driving automation, where robust perception is essential for safe decision-making. Object detection plays a vital role in environmental perception, particularly for collision avoidance. To reduce dependence on multiple sensor technologies, vision-based object detection enables the identification and localization of surrounding vehicles, pedestrians, and traffic infrastructure. However, conventional detectors process frames independently and therefore fail to exploit temporal cues from preceding frames, while existing streaming detectors based on earlier YOLO architectures may struggle to consistently detect all objects in dynamic driving scenarios. To address these limitations, this study introduces StreamYOLO26, a real-time streaming object detection framework built on the latest YOLO26 backbone that exploits temporal information from consecutive frames to improve detection accuracy and consistency. The framework incorporates a C3k2-based Dual Flow Process (DFP) module to fuse temporal features from consecutive frames while retaining the YOLO26 detection pipeline, including its DFL-free detection head, small-target-aware label assignment (sTAL), CIoU and L1 localization losses, binary cross-entropy classification loss, and the Muon with Stochastic Gradient Descent (MuSGD) optimizer. Training follows a two-stage strategy, initially fine-tuning the newly introduced parameters with pretrained components frozen, followed by end-to-end fine-tuning on the Argoverse-HD dataset for direct comparison with existing models. The proposed large model achieves a stream Average Precision (sAP) of 39.2%, with an additional computational cost of 69.18 GFLOPs while reducing latency by 4.2 ms during online evaluation. In contrast, the proposed extra-large model achieves the highest sAP of 42.0% during offline evaluation, demonstrating improved detection accuracy at the expense of increased computational requirements.

Keywords:

Autonomous Vehicle, Streaming Object Detection, Convolution, Attention, Computer Vision.

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: